

ZHEXUAN ZHANG

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Github: github.com/lsCoinred

EDUCATION

Shanghai Tech University, Shanghai, China

Expected graduation July 2027

Bachelor of Engineering in Computer Science and Technology

- Now cumulative GPA: 3.85/4.0, rank 7/171.
- Relevant courses: Natural Language Processing (A+), Introduction to Machine Learning (A), Algorithm and Data Structure (A+), Computer Programming (A+), Compiler (A+), Computer Architecture (A+), Operating System (A+), Linear Algebra (A+).

University of California, Berkeley CA, United States of America

July 2024 - Aug. 2024

Summer Session

- Participated in an intensive academic English and professional communication program.

RESEARCH EXPERIENCE

Shanghai Tech University, VDI Center

Aug. 2025 - Present

Research Assistant. Advised by Prof. Kewei Tu

- **Transformer Grammar** Aug. 2025 - Present
 - Exploring extension of the syntactic language models that simultaneously generate sentences and their syntactic structures, targeting large natural language and coding applications.
 - Working on pre-training the models on large-scale parsed corpora, leveraging distributed training and FlexAttention to improve training efficiency and scalability.
 - Conducting evaluation of the models on downstream language understanding tasks via supervised fine-tuning, analyzing the impact of explicit syntactic modeling.

HONORS & AWARDS

- **Silver Medal**, *The 2025 ICPC Asia Shenyang Regional Contest* 2025
- **Outstanding Teaching Assistant**, *SIST, ShanghaiTech University* 2025
- **Outstanding Student**, *ShanghaiTech University* 2024-2025
- **Outstanding Student**, *ShanghaiTech University* 2023-2024
- **Excellent Individual in Social Practice**, *ShanghaiTech University* 2024

ACTIVITIES & VOLUNTEERING

SI100B: Introduction to Information Science and Technology of *ShanghaiTech*

Dec. - Sept. 2025

Head Teaching Assistant, Advised by Kewei Tu and Xuming He

- Led the deployment and maintenance of the course assignment and examination online judging system.
- Contributed to the design of assignments and exam, and assisted in the tutorial sessions of Python part.

ACM Club of ShanghaiTech

Sept. 2024 - Present

Member & Organizer

- Participated contests, while designing many algorithmic problems for training and contests.

PROJECTS

Content-Based Audio Classification Methods Research

Dec. 2025

Course Project of Introduction to Machine Learning

- Systematically evaluated audio classification paradigms from statistical (HMM-GMM, SVM) to deep learning (1D-CNN, LSTM, ResNet, ViT), analyzing how inductive biases impact modeling of unstructured food chewing sounds.
- Identified that spectral consistency outperforms temporal ordering for food sounds, validating spectral consistency as the primary discriminative feature and informing model selection.
- Designed a lightweight 1D-CNN on MFCC features to capture local spectral correlations and applied K-Fold ensemble to reduce variance, achieving 99.3% accuracy on the Eating Sound Collection dataset.

OAT Compiler

Oct. - Dec. 2024

Course Project of Compiler

- Designed and implemented a full compilation pipeline in OCaml, translating OAT language programs into LLVM IR and x86 assembly, explicitly modeling syntactic structure and semantic constraints.
- Constructed lexer and parser using ocamllex and ocamllyacc, mapping program text to abstract syntax trees (ASTs), gaining strong understanding of formal grammars and structured representations.
- Implemented IR-level optimizations and semantics-preserving program transformations, strengthening understanding of program equivalence, interpretability, and automated code rewriting.

Pintos Operating System

Sept. - Dec. 2025

Course Project of Operating System

- Implemented all core Pintos subsystems (threads, user programs, virtual memory, and file system), building a stable multi-process kernel that passes all test suites.
- Designed and optimized kernel-level concurrency and scheduling mechanisms, including synchronization primitives and priority scheduling, ensuring fairness and throughput under multi-threaded workloads.
- Built a paging-based virtual memory system, together with user-kernel system call and file operation paths, enabling efficient support for compute- and memory-intensive workloads.

SKILLS

- **Languages:** Chinese (Native), English (Fluent, CET-6: 540)
- **Programming:** Proficient in Python, C and C++; familiar with OCaml, MATLAB, and JavaScript
- **Frameworks & Systems:** Familiar with PyTorch, Keras, NumPy, and Librosa; experienced in configuring and running development environments using Linux, Shell scripting, Docker, and Conda;
- **Embedded Systems:** Experienced with STM32, ESP32, Arduino and Raspberry Pi; familiar with SystemVerilog and Simple FPGA Development; capable of basic PCB design.